Theory of Matrices

Video 4.2.1

Definition (Matrices and related terms.)

A **matrix** (plural **matrices**) is a rectangular array of numbers.

The numbers in the array are called **entries** in the matrix.

The **size** of a matrix is given by $m \times n$ where m is the number of rows and n is the number of columns.

row x column
→ always in this order.

The (i, j) -**entry** of a matrix is the number which is in the ith row and jth column of a matrix.

$$\begin{array}{cc} & \text{column 3} \\ & \downarrow \\ \begin{array}{r} \\ \text{row 2} \rightarrow \end{array} & \begin{bmatrix} 1 & 3 & -4 \\ \frac{1}{2} & \pi & 0 \end{bmatrix} \end{array}$$

2x3 matrix
The (2,3) entry is 0.

A sample of the countless uses of matrices

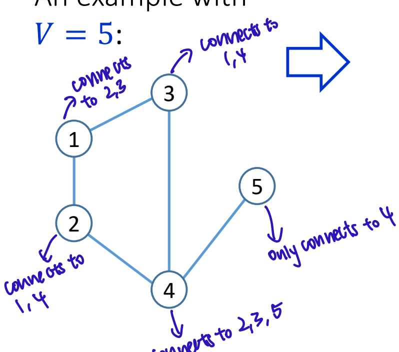
- To represent systems of linear equations
- Represent states, operations, and measurements in quantum physics
- Computer graphics and image compression
- Graph theory: one way of representing a graph is via a matrix.
- Probability theory: Markov chains use matrices encoding probabilities of transistions in state system.
- Coding theory and cryptography

Example: **The adjacency matrix of a graph.**

One was of storing the information in a graph is in an associated matrix called the **adjacency matrix** of the graph.

Take a graph with V vertices.

An example with $V = 5$:



The adjacency matrix is the $\overset{\text{rows}}{V} \times \overset{\text{column}}{V}$ matrix whose (i, j)-entry is $+1$ if vertex i is connected to vertex j and is 0 if the vertices are not connected by an edge.

$$\begin{bmatrix} ? \end{bmatrix}$$

$$\begin{array}{c|ccccc} & 1 & 2 & 3 & 4 & 5 \\ \hline 1 & 0 & 1 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 0 \\ 3 & 1 & 0 & 0 & 1 & 0 \\ 4 & 0 & 1 & 1 & 0 & 1 \\ 5 & 0 & 0 & 0 & 1 & 0 \end{array}$$

Uses of matrices

- To represent systems of linear equations
- Represent states, operations, and measurements in quantum physics
- Computer graphics and image compression
- Graph theory
- Probability theory: Markov chains etc.
- Coding theory and cryptography

Actually, there is an abstract concept that explains most appearances of matrices and most of their properties: matrices are used to represent *linear transformations* between *vector spaces* and they will appear whenever these simple structures are present in the system. These ideas are the main focus of the second course, Linear Algebra II.

To begin

- We'll start by learning some vocabulary and some conventional notation for working with matrices.
- At the beginning this is going to be a long boring list of definitions!

Example

Video 4.2.2

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 0 & -1 \end{bmatrix}$$

The matrix above is a 3×2 matrix.

The $(1,2)$-entry of the matrix is 2.

The $(3,1)$-entry of the matrix is 0.

This way of writing is exactly the same thing: $\rightarrow$ $\begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 0 & -1 \end{pmatrix}$

More Examples

$\begin{pmatrix} 2 & 1 & 0 \end{pmatrix}$ is a 1 x 3 matrix.

$\begin{pmatrix} \sqrt{2} & 3.1 & -2 \\ 3 & \frac{1}{2} & 0 \\ 0 & \pi & 0 \end{pmatrix}$ is a 3 x 3 matrix.

$\begin{pmatrix} 4 \end{pmatrix}$ is a 1 x 1 matrix.

$\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$ is a 3 x 1 matrix.

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This is notation for: the set of all $\overset{\text{row}}{p} \times \overset{\text{column}}{q}$ matrices whose entries are real numbers.

$$\mathbb{M}_{\underbrace{p}_{\text{Number of rows.}} \times \underbrace{q}_{\text{Number of columns.}}}(\mathbb{R})$$

In more advanced subjects like Linear Algebra II, you'll study matrices over different **fields** besides $\mathbb{R}$. For example, the set of $p \times q$ matrices whose entries can be complex numbers is denoted $\mathbb{M}_{p\times q}(\mathbb{C})$.

Definition

A **column matrix** (or a **column vector**) is a matrix with only one column.

A **row matrix** (or a **row vector**) is a matrix with only one row.

Example

$\begin{pmatrix} 2 & 1 & 0 \end{pmatrix}$ is a row matrix.

$\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$ is a column matrix.

(4) is a row matrix as well as a column matrix.

Is this different to $\mathbb{R}^n$?

The elements of $\mathbb{M}_{n\times 1}$ and $\mathbb{M}_{1\times n}$ seem very similar to the elements of $\mathbb{R}^n$. There are all just lists of n real numbers, but presented differently.

For example, consider $n = 3$. We actually have three closely related but different things:

A vector:	A row vector:	A column vector:
$\vec{x} = (a, b, c) \in \mathbb{R}^3$	$\boldsymbol{X} = [a \;\; b \;\; c] \in \mathbb{M}_{1\times 3}$	$\boldsymbol{X}^T = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \in \mathbb{M}_{1\times 3}$

(We will explain this notation later this lecture.)

These three sets are different but **isomorphic**.

We will discuss “isomorphism” later in this course, and much more seriously in *Linear Algebra II*. But it means that they are basically the same mathematical structure but maybe they *look* different.

For example consider $\mathbb{R}^n$ and $\mathbb{M}_{1\times n}$. There are obvious functions going from each of the two sets to the other one. These functions set up a correspondence (a “bijection”) between the sets. They tell which element in the second set an element in the first set corresponds to.

$$\mathbb{R}^n \rightleftarrows \mathbb{M}_{1\times n}$$

$$\vec{x} = (a, b, c) \qquad \boldsymbol{X} = \begin{bmatrix} a & b & c \end{bmatrix}$$

$$\mathbb{R}^n \rightleftarrows \mathbb{M}_{1\times n}$$

$$\vec{x} = (a, b, c) \qquad \boldsymbol{X} = \begin{bmatrix} a & b & c \end{bmatrix}$$

In this course we will usually be a bit casual and work as if these are the same spaces. In Linear Algebra II we will be more careful.

Notations

Video 4.2.3

The specific entries of an $m \times n$ matrix are often written using the following notation a_{ij}:

$$\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

If we have defined the collection of numbers a_{ij} first then we can define the matrix $\boldsymbol{A}$ by writing

$$\boldsymbol{A} = [a_{ij}].$$

Sometimes we write

$$\boldsymbol{A} = [a_{ij}]_{m\times n}$$

when we want to emphasize the shape of the matrix.

Examples of definitions of matrices

1. $\boldsymbol{A} = (a_{ij})_{2\times 3}$ where $a_{ij} = i + j$

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$$

2. $\boldsymbol{B} = (b_{ij})_{3\times 2}$ where $b_{ij} = \begin{cases} 1 & \text{if } i+j \text{ is even.} \\ -1 & \text{if } i+j \text{ is odd.} \end{cases}$

$$B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \\ 1 & -1 \end{bmatrix}$$

Exercise:

Write down the matrix defined by the following formulae

$$\boldsymbol{A} = [a_{ij}]_{2\times 3}, \qquad a_{ij} = i^2 - j^2 - i - j + 1.$$

Solution

$$= \begin{bmatrix} 1^2 - 1^2 - 1 - 1 + 1 & 1^2 - 2^2 - 1 - 2 + 1 & 1^2 - 3^2 - 1 - 3 + 1 \\ 2^2 - 1^2 - 2 - 1 + 1 & 2^2 - 2^2 - 2 - 2 + 1 & 2^2 - 3^2 - 2 - 3 + 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & -5 & -11 \\ 1 & -3 & -9 \end{bmatrix}$$

$2^2 - (2)^2 - 2 - 2 + 1$

Notation

We'll sometimes also use the following standard variation on the notation.

If $\boldsymbol{A}$ is a matrix, then $\boldsymbol{A}_{ij}$ denotes the entry in row i and column j of the matrix $\boldsymbol{A}$.

Example: $\boldsymbol{A} = \begin{bmatrix} 2 & 4 \\ -6 & -8 \\ 10 & 12 \end{bmatrix}$. Then $\boldsymbol{A}_{31} = 10$.

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Definition

A matrix is called a **square matrix** if it has the same number of rows and columns.

In particular, an $n \times n$ matrix is called a square matrix of order n.

e.g. (4) $\begin{pmatrix} 0 & 4 \\ 1 & 2 \end{pmatrix}$ $\begin{pmatrix} 1 & 2 & 3 \\ -1 & 3 & 2 \\ 0 & 0 & 2 \end{pmatrix}$ $\begin{pmatrix} 2 & 1 & 6 & 2 \\ 0 & 3 & 9 & -1 \\ 1 & 0 & 0 & 0 \\ -2 & 3 & 0 & 1 \end{pmatrix}$

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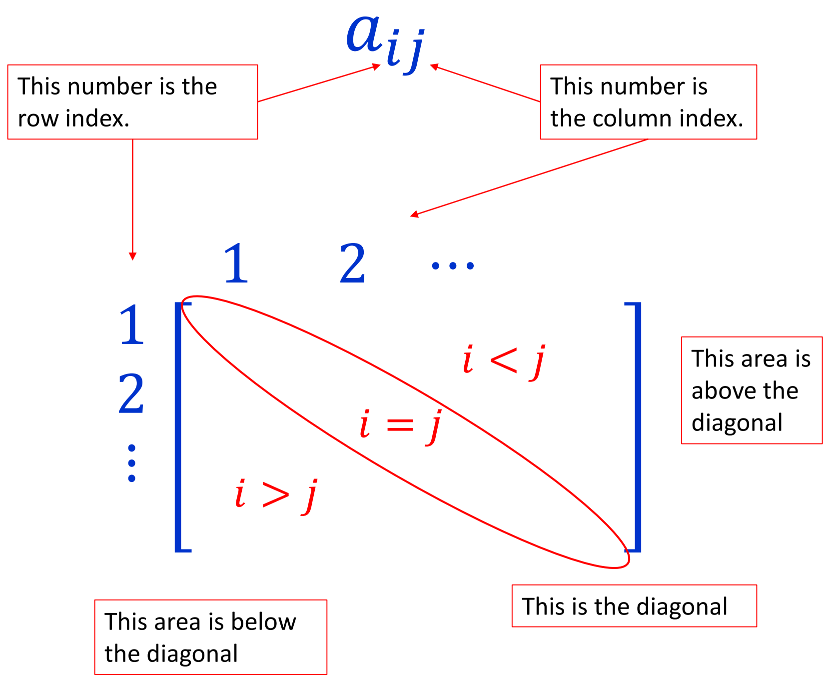
Definition

Given a square matrix $\boldsymbol{A} = (a_{ij})$ of order $n > 0$, the **diagonal** of $\boldsymbol{A}$ is the sequence of entries

$$a_{11},\ a_{22},\ \ldots,\ a_{nn}.$$

The entries a_{ii} are called the **diagonal entries** while a_{ij} , $i \neq j$, are called **non-diagonal entries**.

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$$



<u>Definition</u>

A square matrix is called a **diagonal matrix** if all its non-diagonal entries are zero, i.e.

$\boldsymbol{A} = (a_{ij})_{n \times n}$ is diagonal if $a_{ij} = 0$ whenever $i \neq j$.

e.g. (4) $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ $\begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

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Definition

A diagonal matrix is called a **scalar matrix** if all its diagonal entries are the same.

$\boldsymbol{A} = (a_{ij})_{n \times n}$ is scalar if for some constant $c \in \mathbb{R}$

$$a_{ij} = \begin{cases} c & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

e.g. (4) $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ $\begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}$ → 3 times of a 4x4 identity matrix.

Definition

A diagonal matrix is called the **identity** matrix if all its diagonal entries are 1.

We use $\boldsymbol{I}_n$ to denote the identity matrix of order n.

Sometimes we write $\boldsymbol{I}$ instead of $\boldsymbol{I}_n$ when there is no danger of confusion.

e.g.

$$\boldsymbol{I}_1 = (1) \quad \boldsymbol{I}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \boldsymbol{I}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \boldsymbol{I}_4 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Definition

A matrix with all entries equal to zero is called the **zero matrix**.

We use $\mathbf{0}_{m\times n}$ to denote the zero matrix of size $m \times n$.

Sometimes we write $\mathbf{0}$ instead of $\mathbf{0}_{m\times n}$ when there is no danger of confusion.

e.g. $\mathbf{0}_{1\times 1} = (0)$ $\mathbf{0}_{2\times 4} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ $\mathbf{0}_{4\times 3} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

Definition

A square matrix (a_{ij}) is called a **symmetric matrix** if
$$a_{ij} = a_{ji} \text{ for all } i, j.$$

*similar to even functions.

e.g. (4) $\begin{pmatrix} 0 & 4 \\ 4 & 2 \end{pmatrix}$ $\begin{pmatrix} 1 & -1 & 0 \\ -1 & 3 & 2 \\ 0 & 2 & 2 \end{pmatrix}$ $\begin{pmatrix} 2 & 1 & 6 & -2 \\ 1 & 3 & 0 & -1 \\ 6 & 0 & 0 & 0 \\ -2 & -1 & 0 & 1 \end{pmatrix}$

Definition

A square matrix (a_{ij}) is called **upper triangular** if

$$a_{ij} = 0 \text{ for all } i > j.$$

A square matrix (a_{ij}) is called **lower triangular** if

$$a_{ij} = 0 \text{ for all } i < j.$$

Both upper and lower triangular matrices are called **triangular matrices.**

e.g. (4) $\begin{pmatrix} 0 & 4 \\ 0 & 2 \end{pmatrix}$ $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 3 & 2 \\ 0 & 0 & 2 \end{pmatrix}$ $\begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ -2 & 3 & 0 & 1 \end{pmatrix}$

Upper triangular. Upper triangular. Lower triangular.

Definition

Two matrices are said to be equal if they have the same size and their corresponding entries are equal.

Let $\boldsymbol{A} = (a_{ij})_{m \times n}$ and $\boldsymbol{B} = (b_{ij})_{p \times q}$.

Then $\boldsymbol{A}$ is equal to $\boldsymbol{B}$ if

$$m = p, \ n = q \text{ and } a_{ij} = b_{ij} \text{ for all } i, j.$$

Example

$$\boldsymbol{A} = \begin{pmatrix} 1 & x \\ 2 & 4 \end{pmatrix} \quad \boldsymbol{B} = \begin{pmatrix} 1 & -1 \\ 2 & 4 \end{pmatrix} \quad \boldsymbol{C} = \begin{pmatrix} 1 & -1 & 0 \\ 2 & 4 & 0 \end{pmatrix}$$

where x is a constant.

1. $\boldsymbol{A} = \boldsymbol{B}$ if and only if $x = -1$.
2. $\boldsymbol{A} \neq \boldsymbol{C}$ for all value of x.

Definition (Trace of a matrix).

Video 4.2.4

The **trace** is an important function defined on *square* matrices

$$tr: \{square\ matrices\} \to \mathbb{R}.$$

It is defined to be the sum of the diagonal elements.

In symbols: Given a matrix $(a_{ij})_{n \times n}$ then: $tr(A) = \sum_{i=1}^{n} a_{ii}$.

Example:

$$tr\left(\begin{bmatrix} 1 & 2 & 3 & 4 \\ -1 & 0 & 1 & 2 \\ -3 & -2 & -1 & 0 \\ -5 & 0 & 5 & 10 \end{bmatrix}\right) = 1 + 0 + (-1) + 10 = 10 \in \mathbb{R}.$$

This functions is important because it has nice properties, such as "cyclic invariance": $tr(AB) = tr(BA)$. These properties will be explored in the tutorial.

Definition (Transpose).

The **transpose** is another important function defined on matrices of any shape. The transpose of a matrix $\boldsymbol{M}$ is denoted $\boldsymbol{M}^T$:

$$transpose\colon \mathbb{M}_{p\times q}(\mathbb{R}) \to \mathbb{M}_{q\times p}(\mathbb{R})$$

Note:
- the (# of rows) of $\boldsymbol{M}$ becomes the (# of columns) of the transpose $\boldsymbol{M}^T$.
- the (# of columns) of $\boldsymbol{M}$ becomes the (# of rows) of the transpose $\boldsymbol{M}^T$.

The definition is:

If $\boldsymbol{A} = \left(a_{ij}\right)_{p\times q}$ then $\boldsymbol{A}^T$ is defined to be the matrix $\boldsymbol{B} = \left(b_{ij}\right)_{q\times p}$ where

$$b_{ij} = a_{ji}.$$

Examples of transpose

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ -1 & 0 & 1 & 2 \\ -3 & -2 & -1 & 0 \\ -5 & 0 & 5 & 10 \end{bmatrix}^T = \begin{bmatrix} 1 & -1 & -3 & -5 \\ 2 & 0 & -2 & 0 \\ 3 & 1 & -1 & 5 \\ 4 & 2 & 0 & 10 \end{bmatrix}$$

Ways of thinking about this:
- Rows become columns and columns become rows
- You are "reflecting" the matrix over the diagonal

$$M_{1\times n}(\mathbb{R}) \underset{\text{take the transpose}}{\overset{\text{take the transpose}}{\rightleftarrows}} M_{n\times 1}(\mathbb{R}) \Rightarrow \text{isomorphism}$$

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \qquad \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Examples of transpose (cont)

$$\begin{bmatrix} 3 & 1 & -1 & 5 \\ 4 & 2 & 0 & 10 \end{bmatrix}^T = \begin{bmatrix} 3 & 4 \\ 1 & 2 \\ -1 & 0 \\ 5 & 10 \end{bmatrix} \quad 2\times 4 \to 4\times 2$$

$$\begin{bmatrix} 3 \\ 1 \\ -1 \\ 5 \end{bmatrix}^T = \begin{bmatrix} 3 & 1 & -1 & 5 \end{bmatrix} \quad 4\times 1 \to 1\times 4$$

$$[10]^T = [10]$$

Transpose is important because it is a simple and natural operation you can perform on a matrix.

For example: it is easy to see that a matrix is symmetric if and only if it is equal to its transpose.

Video 4.2.5

Matrix algebra

We introduced matrices as a simple notation to organize our calculations of systems of linear equations.

But there is a wonderful thing about mathematics – often when you introduce something for some particular reason, it takes on a life of its own, and goes to places you never imagined.

The crucial next step in the matrix story is to understand that they are more than just a table, but they have ***algebraic operations***, just like number systems like $\mathbb{R}$ and $\mathbb{C}$.

Same no. of rows and columns.

The three basic algebraic operations on matrices:

Matrix operation #1: Matrix addition. When matrices have the same shape you can add them together.

$$\begin{bmatrix} 1 & 0.1 & 0 & 4 \\ -2 & 2 & 10 & 0 \\ 2 & 1 & -1 & -2 \end{bmatrix} + \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0.6 & -8 & 3 & 2 \\ 100 & 1 & 1 & -2 \end{bmatrix} = \begin{bmatrix} 2 & 0.1 & -1 & 4 \\ -1.4 & -6 & 13 & 2 \\ 102 & 2 & 0 & -4 \end{bmatrix}$$

Matrix operation #2: Multiplication by a scalar. Any matrix $\boldsymbol{M}$ can be multiplied by any scalar $c \in \mathbb{R}$ to get a new matrix $c\boldsymbol{M}$.

$$\frac{2}{5}\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} = \begin{bmatrix} 2/5 & 4/5 \\ 6/5 & 8/5 \\ 2 & 12/5 \end{bmatrix}.$$

Matrix operation #3: Matrix multiplication.

We will focus on the details of this surprising operation carefully.

The main things we need to do in this lecture:

- Carefully define and understand these basic operations, spending a lot of time on the most interesting one, matrix multiplication.
- Write down (and prove a few of) the algebraic rules that these operations satisfy.

Operation #1 of 3: Matrix addition.

Definition (Matrix addition).

Take two matrices of the same shape $\boldsymbol{M}_1, \boldsymbol{M}_2 \in \mathbb{M}_{m\times n}(\mathbb{R})$.
The sum of these two matrices

$$\boldsymbol{M}_1 + \boldsymbol{M}_2 \in \mathbb{M}_{m\times n}(\mathbb{R})$$

is defined to be the matrix with the same shape whose (i, j)-entry is the sum of the (i, j)-entry of $\boldsymbol{M}_1$ plus the (i, j)-entry of $\boldsymbol{M}_2$.

Alternatively, writing this using formulae:

If $\boldsymbol{A} = \left[a_{ij}\right]_{m\times n}$ and $\boldsymbol{B} = \left[b_{ij}\right]_{m\times n}$ then $\boldsymbol{A} + \boldsymbol{B}$ equals the matrix $\boldsymbol{C} = \left[c_{ij}\right]_{m\times n}$ where

$$c_{ij} = a_{ij} + b_{ij}.$$

For example:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} + \begin{bmatrix} -2 & 4 \\ -8 & 16 \\ -32 & 64 \end{bmatrix} = \begin{bmatrix} -1 & 6 \\ -5 & 20 \\ -27 & 70 \end{bmatrix}.$$

To find the entry of some position in the result of addition…

Take the entries from the corresponding positions in the summands and add them together.

In this example: $5 + 32 = -27$.

Operation #2 of 3: Scalar multiplication.

Definition (Scalar multiplication of a matrix).

Take a scalar $r \in \mathbb{R}$.
Take a matrix $\boldsymbol{M} \in \mathbb{M}_{m\times n}(\mathbb{R})$.
The matrix that results when you multiply this matrix $\boldsymbol{M}$ by the scalar r, denoted

$$r\boldsymbol{M} \in \mathbb{M}_{m\times n}(\mathbb{R})$$

is defined to be the matrix with the same shape as $\boldsymbol{M}$ whose (i, j)-entry is the the result of multiplying the (i, j)-entry of $\boldsymbol{M}$ by the scalar r.

Alternatively, writing this using formulae:

If $\boldsymbol{A} = [a_{ij}]_{m\times n}$ and $r \in \mathbb{R}$ then $r\boldsymbol{A}$ equals the matrix $\boldsymbol{C} = [c_{ij}]_{m\times n}$ where

$$c_{ij} = ra_{ij}.$$

For example:

$$3\begin{bmatrix} 9 & -8 & 7 \\ -6 & 5 & -4 \\ 3 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 27 & -24 & 21 \\ -18 & 15 & -12 \\ 9 & -6 & 3 \end{bmatrix}$$

To find the entry of some position in the result of scalar multiplication…

… take the corresponding entry of the original matrix and multiply it by the scalar. In this case: $(3) * (-4) = -12$.

You might ask:

"Where is matrix subtraction? Can't we define that? Shouldn't that be one of the basic operations?"

Yes we can define that in the obvious way:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} - \begin{bmatrix} 10 & 8 & 6 \\ 4 & 2 & 0 \\ -2 & -4 & -6 \end{bmatrix}$$

$$= \begin{bmatrix} 1-10 & 2-8 & 3-6 \\ 4-4 & 5-2 & 6-0 \\ 7-(-2) & 8-(-4) & 9-(-6) \end{bmatrix} = \begin{bmatrix} -9 & -6 & -3 \\ 0 & 3 & 6 \\ 9 & 12 & 15 \end{bmatrix}$$

Question

Why didn't I list matrix subtraction as the of the basic matrix operations?

Answer

Because it can be obtained as a combination of the other two operations, so it shouldn't be considered "basic". It would be redundant to include it.

How? Can we do that?

$$\boldsymbol{M} - \boldsymbol{N} = \boldsymbol{M} + (-1)\boldsymbol{N}.$$

Comment

The notation $-\boldsymbol{M}$ is defined to mean $(-1)\boldsymbol{M}$.

Video 4.2.6

Before we tackle the final algebraic operation (matrix multiplication) let's stop and think about what should be the list of rules for matrix algebra which just use the two operations we have defined so far.

Theorem:

Let $\boldsymbol{M}, \boldsymbol{N}, \boldsymbol{P} \in \mathbb{M}_{m\times n}(\mathbb{R})$ be matrices of the same shape.

Recall that $\mathbf{0} \in \mathbb{M}_{m\times n}(\mathbb{R})$ denotes the zero matrix.

Let $r, s \in \mathbb{R}$ be real numbers.

Then:

- $(\boldsymbol{M} + \boldsymbol{N}) + \boldsymbol{P} = \boldsymbol{M} + (\boldsymbol{N} + \boldsymbol{P})$
- $\boldsymbol{M} + \boldsymbol{N} = \boldsymbol{N} + \boldsymbol{M}$
- $\mathbf{0} + \boldsymbol{M} = \boldsymbol{M} + \mathbf{0} = \boldsymbol{M}$

And

- $r(s\boldsymbol{M}) = (rs)\boldsymbol{M}$
- $1\boldsymbol{M} = \boldsymbol{M}$
- $0\boldsymbol{M} = \mathbf{0}$

And

- $(r + s)\boldsymbol{M} = r\boldsymbol{M} + s\boldsymbol{M}$
- $r(\boldsymbol{M} + \boldsymbol{N}) = r\boldsymbol{M} + r\boldsymbol{N}$

Theorem:

Let $\boldsymbol{M}, \boldsymbol{N}, \boldsymbol{P} \in \mathbb{M}_{m \times n}(\mathbb{R})$ be matrices of the same shape.

Recall that $\mathbf{0} \in \mathbb{M}_{m \times n}(\mathbb{R})$ denotes the zero matrix.

Let $r, s \in \mathbb{R}$ be real numbers.

Then:

- $(\boldsymbol{M} + \boldsymbol{N}) + \boldsymbol{P} = \boldsymbol{M} + (\boldsymbol{N} + \boldsymbol{P})$ Associativity of matrix addition.
- $\boldsymbol{M} + \boldsymbol{N} = \boldsymbol{N} + \boldsymbol{M}$ Commutativity of matrix addition.
- $\mathbf{0} + \boldsymbol{M} = \boldsymbol{M} + \mathbf{0} = \boldsymbol{M}$ Adding zero matrix has zero effect.

And

- $r(s\boldsymbol{M}) = (rs)\boldsymbol{M}$
- $1\boldsymbol{M} = \boldsymbol{M}$
- $0\boldsymbol{M} = \mathbf{0}$ zero matrix

The first three axioms only talk about matrix addition by itself.

And

- $(r + s)\boldsymbol{M} = r\boldsymbol{M} + s\boldsymbol{M}$
- $r(\boldsymbol{M} + \boldsymbol{N}) = r\boldsymbol{M} + r\boldsymbol{N}$

Theorem:

Let $\boldsymbol{M}, \boldsymbol{N}, \boldsymbol{P} \in \mathbb{M}_{m \times n}(\mathbb{R})$ be matrices of the same shape.

Recall that $\mathbf{0} \in \mathbb{M}_{m \times n}(\mathbb{R})$ denotes the zero matrix.

Let $r, s \in \mathbb{R}$ be real numbers.

Then:

- $(\boldsymbol{M} + \boldsymbol{N}) + \boldsymbol{P} = \boldsymbol{M} + (\boldsymbol{N} + \boldsymbol{P})$
- $\boldsymbol{M} + \boldsymbol{N} = \boldsymbol{N} + \boldsymbol{M}$
- $\mathbf{0} + \boldsymbol{M} = \boldsymbol{M} + \mathbf{0} = \boldsymbol{M}$

And

- $r(s\boldsymbol{M}) = (rs)\boldsymbol{M}$
- $1\boldsymbol{M} = \boldsymbol{M}$
- $0\boldsymbol{M} = \mathbf{0}$ zero matrix

The next three axioms talk about scalar multiplication by itself.

And

- $(r + s)\boldsymbol{M} = r\boldsymbol{M} + s\boldsymbol{M}$
- $r(\boldsymbol{M} + \boldsymbol{N}) = r\boldsymbol{M} + r\boldsymbol{N}$

The final two axioms mix matrix addition with scalar multiplication. These are called “distributive” rules. There are two types depending on what you are adding.

<u>Theorem:</u>

Let $\boldsymbol{M}, \boldsymbol{N}, \boldsymbol{P} \in \mathbb{M}_{m\times n}(\mathbb{R})$ be matrices of the same shape.

Recall that $\mathbf{0} \in \mathbb{M}_{m\times n}(\mathbb{R})$ denotes the zero matrix.

Let $r, s \in \mathbb{R}$ be real numbers.

Then:

- $(\boldsymbol{M}+\boldsymbol{N})+\boldsymbol{P} = \boldsymbol{M}+(\boldsymbol{N}+\boldsymbol{P})$
- $\boldsymbol{M}+\boldsymbol{N} = \boldsymbol{N}+\boldsymbol{M}$
- $\mathbf{0}+\boldsymbol{M} = \boldsymbol{M}+\mathbf{0} = \boldsymbol{M}$

Question:
I didn't include the rule $\boldsymbol{M}-\boldsymbol{M}=\mathbf{0}$ in this list. Why?

not a fundamental operation.

And

- $r(s\boldsymbol{M}) = (rs)\boldsymbol{M}$
- $1\boldsymbol{M} = \boldsymbol{M}$
- $0\boldsymbol{M} = \mathbf{0}$

And

- $(r+s)\boldsymbol{M} = r\boldsymbol{M}+s\boldsymbol{M}$
- $r(\boldsymbol{M}+\boldsymbol{N}) = r\boldsymbol{M}+r\boldsymbol{N}$

<u>Problem</u>

Give a short proof that the rule

$$\boldsymbol{M}-\boldsymbol{M}=\mathbf{0}$$

for any matrix $\boldsymbol{M}$, follows from the axioms listed in the theorem.

<u>Solution</u>

$$
\begin{aligned}
\boldsymbol{M}-\boldsymbol{M} &= \boldsymbol{M}+(-1)\boldsymbol{M} && \text{(Definition of } \boldsymbol{M}-\boldsymbol{M}\text{)}\\
&= 1M + (-1)M && \text{(Scalar multiplication by 1)}\\
&= [1+(-1)]M && \text{(Distributive law)}\\
&= (0)M && \text{(Calculation in } \mathbb{R}\text{)}\\
&= 0 && \text{(Scalar multiplication by 0)}
\end{aligned}
$$

$\square$

Proofs of these properties.

The proofs of all these rules are very simple and mechanical. They always come back to some similar property about the real numbers $\mathbb{R}$.

Let's randomly choose one of the rules and prove it. (There will be a few more exercises on this week's tutorial.)

Video 4.2.7

Proof of the distributive rule: $r(\boldsymbol{M}+\boldsymbol{N}) = r\boldsymbol{M} + r\boldsymbol{N}$.

Let $\boldsymbol{M}$ and $\boldsymbol{N}$ be arbitrary matrices of the same shape $\boldsymbol{M}, \boldsymbol{N} \in \mathbb{M}_{a\times b}$.

Let $r \in \mathbb{R}$.

Assume that the entries of the two matrices are given in this way $\boldsymbol{M} = [m_{ij}]_{a\times b}$ and $\boldsymbol{N} = [n_{ij}]_{a\times b}$.

To prove two matrices are equal we need to prove that all of the corresponding entries are equal.

So we need to show the entries of the two sides of

$$r(\boldsymbol{M}+\boldsymbol{N}) = r\boldsymbol{M} + r\boldsymbol{N}$$

match up:

$$\text{The } (i,j) \text{ entry of } r(\boldsymbol{M}+\boldsymbol{N}) = \text{The } (i,j) \text{ entry of } (r\boldsymbol{M} + r\boldsymbol{N})$$

Proof (Continued):

So let's calculate:

The (i, j) entry of $r(\boldsymbol{M}+\boldsymbol{N})$

$= r * \big(\text{The } (i, j) \text{ entry of } (\boldsymbol{M}+\boldsymbol{N})\big)$ This is the definition of scalar multiplication.

$= r * \big(m_{ij} + n_{ij}\big)$ This is the definition of matrix addition.

$= rm_{ij} + rn_{ij}$ This equation is true in $\mathbb{R}$.

$= \text{The } (i, j) \text{ entry of } (r\boldsymbol{M}) + \text{The } (i, j) \text{ entry of } (r\boldsymbol{N})$

This is two uses of the definition of scalar multiplication.

$= \big(\text{The } (i, j) \text{ entry of } (r\boldsymbol{M}+r\boldsymbol{N})\big)$

This is the definition of matrix addition.

$\square$

Video 4.2.8

A short digression:

A surprising comparison with $\mathbb{R}^n$ and a hint of deeper mathematical ideas.

Let's compare what we have learnt about matrices so far with another mathematical structure $\mathbb{R}^n$.

(No, this section isn't examinable in MH1200. MH1201 is a different story however…)

	$\mathbb{R}^{\boldsymbol{n}}$	$\mathbb{M}_{\boldsymbol{p}\times\boldsymbol{q}}(\mathbb{R})$
What are the elements?	n-tuples of real numbers.	$p \times q$-matrices of real numbers.
What are the algebraic operations?	• Vector addition. • Scalar multiplication.	• Matrix addition. • Scalar multiplication. • (Matrix multiplication, but forget that for this comparison.)
What rules do these operations satisfy?	• Addition is associative. • Addition is commutative. • There is a special zero vector $\vec{0}$. • The rule: $r\big(s(\vec{v})\big) = (rs)\vec{v}$. • Multiplying a vector by $1 \in \mathbb{R}$ does nothing to the vector. • Multiplying a vector by $0 \in \mathbb{R}$ turns it into the zero vector. • $r(\vec{v} + \vec{w}) = r\vec{v} + r\vec{w}$. • $(r+s)\vec{v} = r\vec{v} + s\vec{w}$.	• Addition is associative. • Addition is commutative. • There is a special zero matrix $\mathbf{0}$. • The rule: $r\big(s(\boldsymbol{M})\big) = (rs)\boldsymbol{M}$. • Multiplying a matrix by $1 \in \mathbb{R}$ does nothing to the matrix. • Multiplying a matrix by $0 \in \mathbb{R}$ turns it into the zero matrix. • $r(\boldsymbol{M} + \boldsymbol{N}) = r\boldsymbol{M} + r\boldsymbol{N}$. • $(r+s)\boldsymbol{M} = r\boldsymbol{M} + s\boldsymbol{M}$.

What do you notice?

What do you notice?

They seem to be exactly the same structure only the names of the elements are different!

Mathematicians love this sort of thing – seeing the same structure in different places but disguised. The mathematical word for this is **isomorphism**.

So mathematicians invented a new concept which simultaneously generalizes $\mathbb{R}^n$ and $\mathbb{M}_{p\times q}(\mathbb{R})$ and many other examples: ***(Abstract) Vector Space***.

These are the objects you'll focus on in Linear Algebra II. Maybe go Google them.

The category of vector spaces over $\mathbb{R}$.

$\mathbb{M}_{p\times q}(\mathbb{R})$

$\mathbb{R}^n$

Polynomials with $\mathbb{R}$ coefficients of degree $\leq d$

Continuous functions $\mathbb{R} \to \mathbb{R}$

Sequences of $\mathbb{R}$ numbers: $\{s_i\}_{i=1}^{\infty}$

……

Linear transformations between vector spaces.